VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

MILESTONE 2

Image Processing & Defect Detection

Week 3 & 4

Milestone Progress Document

1. Project Overview

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System is an AI-powered manufacturing quality-inspection platform intended to detect product defects from images, identify quality issues, classify defect types, and provide production analytics. The platform is designed to reduce manual inspection effort, improve product quality, minimize manufacturing defects, and increase production efficiency.

2. Milestone 2 Objective

Develop the computer-vision inspection pipeline that prepares product images, performs image-quality analysis, trains or integrates defect-detection models, generates predictions, and exposes inspection insights through the monitoring workflow.

3. Scope of Work

- Implemented image preprocessing pipelines.
- Prepared image-quality analysis workflows and reports.
- Built image analytics workflows for inspection.
- Trained or integrated defect-detection models as specified for the project.
- Generated defect predictions from product images.
- Integrated inspection results into monitoring dashboards.

4. Planned System Modules

- Image Acquisition
- Image Processing
- Defect Detection

5. Key Deliverables

- Image preprocessing pipeline
- Image-quality analysis workflow
- Defect-detection model integration
- Defect prediction engine
- Inspection monitoring workflow
- Inspection report generation

6. Expected / Recorded Outcomes

- Implemented image processing and defect-detection workflows.
- Built an AI-powered computer-vision inspection pipeline.
- Developed understanding of image processing and anomaly-detection concepts.
- Generated defect-detection insights from inspection images.

7. Technology Stack

Backend: Python (FastAPI) | Frontend: React.js / Next.js | Database: PostgreSQL / MongoDB

AI & Computer Vision: OpenCV, TensorFlow, PyTorch, YOLO, CNN models, Scikit-learn, NumPy and Pandas.

Deployment & Development: Docker, Docker Compose, AWS/Azure, Git/GitHub, VS Code and Postman.

8. Performance & Evaluation Focus

The project specification identifies defect-detection accuracy, precision, recall, F1-score, mAP, inspection automation rate, defect identification accuracy, false-defect detection rate, image-processing speed, inspection response time, and concurrent inspection capability as key metrics.

9. Challenges / Notes

This milestone document is based on the supplied VisionInspect AI project specification. Where implementation evidence, measured model metrics, screenshots, deployment URLs, or issue logs were not included in that source, this document does not invent those details. Actual measured results and screenshots can be added to the GitHub copy before submission.

10. Next Milestone / Continuation

Add defect categorization, severity scoring, quality assessment, production reporting, and manufacturing analytics.

Conclusion

Milestone 2 establishes the planned progression of the VisionInspect AI system toward an end-to-end manufacturing defect detection and quality inspection platform. The work in this milestone provides the foundation for the next stage of implementation and final demonstration.